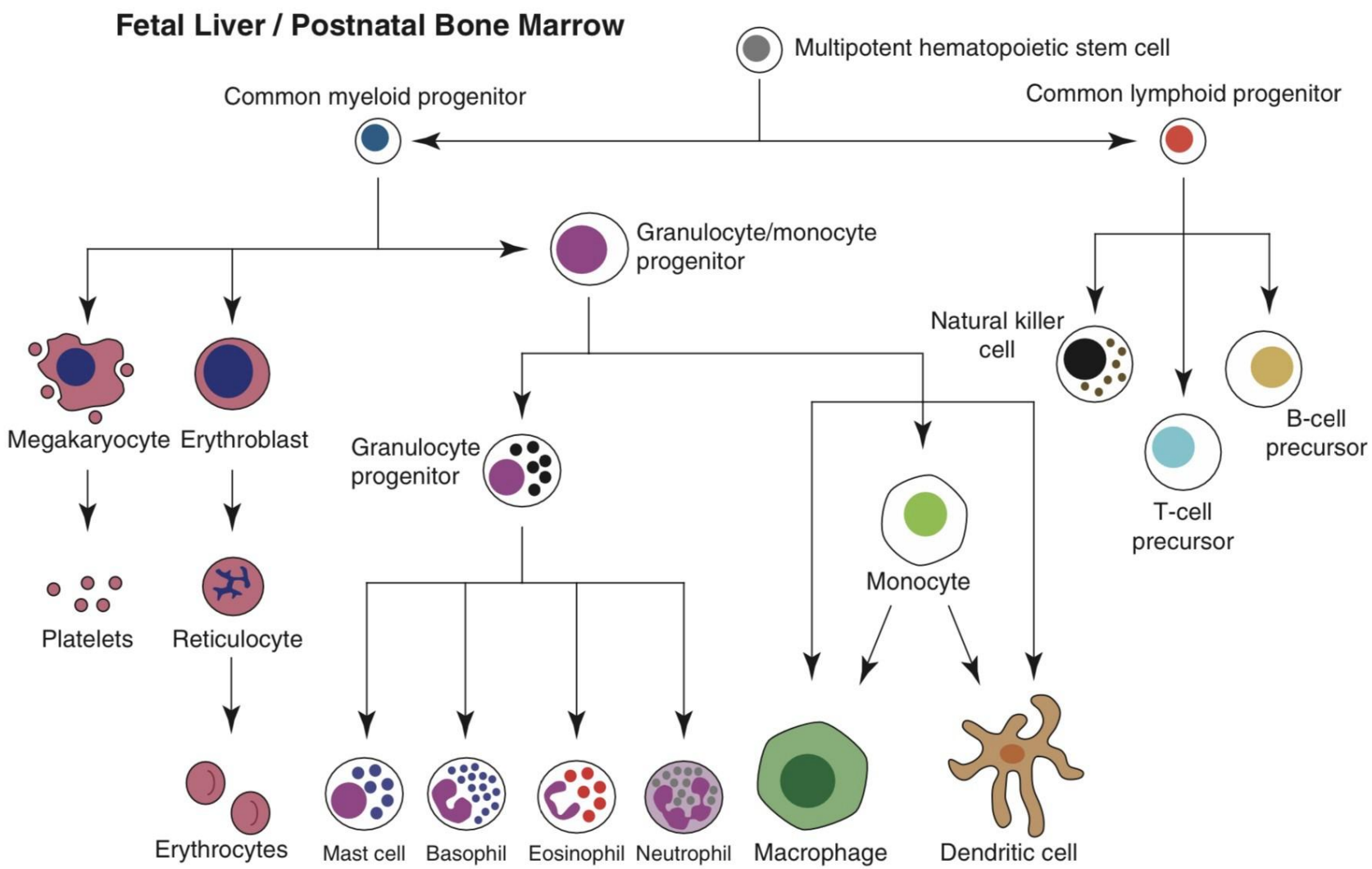
Lecture

BLOOD

- Functions of Blood.
- Composition of Blood.
- Red Blood Cells (Erythrocytes).
- White Blood Cells: Granulocytes.
- White Blood Cells: Agranulocytes.
- Other Immune Cells.
- Platelets (Thrombocytes).


Fetal Liver / Postnatal Bone Marrow
Multipotent hematopoietic stem cell
Common myeloid progenitor
Common lymphoid progenitor
Granulocyte/monocyte progenitor
Megakaryocyte
Erythroblast
Platelets
Reticulocyte
Erythrocytes
Granulocyte progenitor
Mast cell
Basophil
Eosinophil
Neutrophil
Monocyte
Macrophage
Dendritic cell
Natural killer cell
T-cell precursor
B-cell precursor

Functions of Blood

1. **Transport**: Moves oxygen, nutrients, waste, and hormones around the body.
2. **Regulation**: Keeps pH, temperature, and water levels stable.
3. **Protection**: Stops bleeding through clotting and fights infections with white blood cells.
4. **Waste Removal**: Carries waste to organs for removal.
5. **Homeostasis**: Maintains balance in the body's internal environment.

Composition of Blood

- **Cells (~45%):** Produced in the bone marrow. These include:
  - **Red blood cells (erythrocytes):** ~5 × 10^6/mL
  - **White blood cells (leucocytes):** ~2 × 10^3/mL
  - **Platelets (thrombocytes):** ~2 × 10^5/mL

- **Plasma (~55%):** Composed mainly of water (92%) and contains proteins, lipids, salts, and glucose.

Red Blood Cells (Erythrocytes)

Function:

Transport oxygen and carbon dioxide via hemoglobin.

Structure:

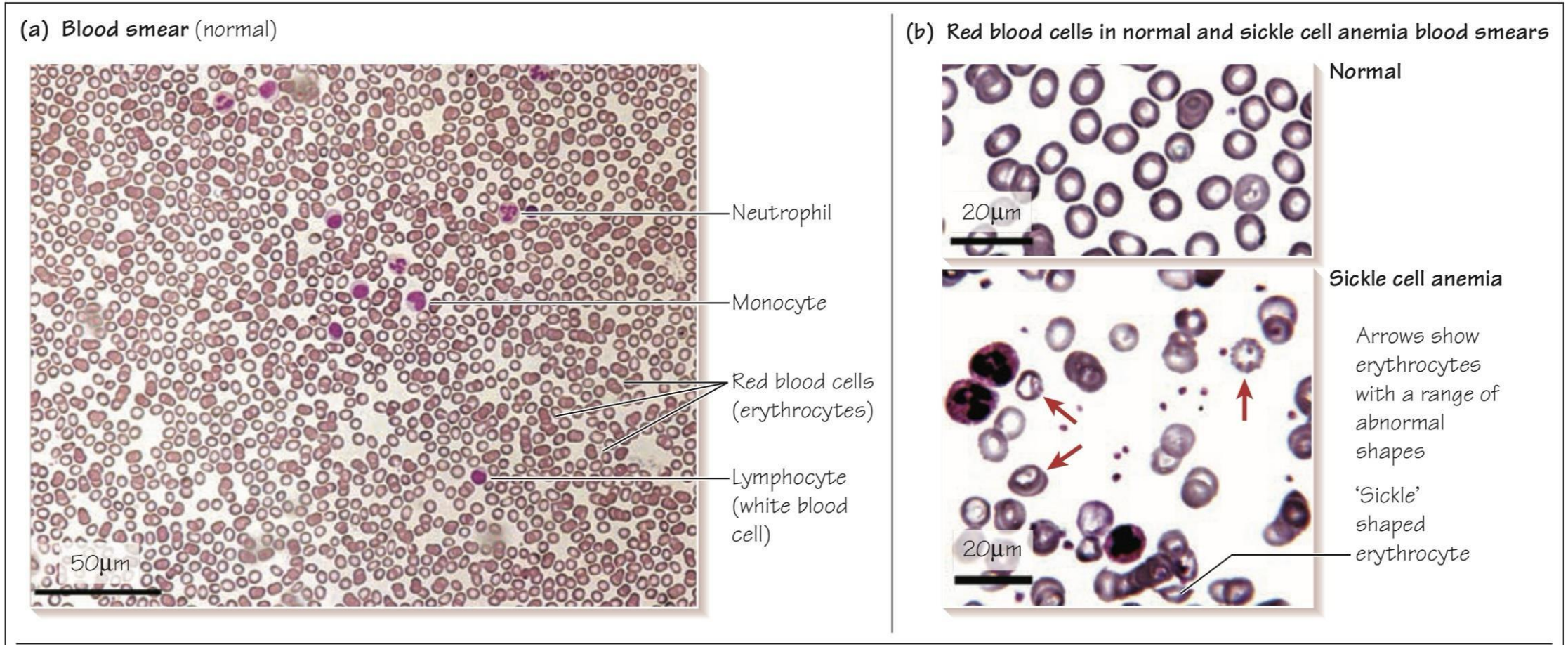
- ~7 μm in diameter, biconcave, and eosinophilic due to high protein content.
- Lose their nuclei, making them flexible to pass through capillaries.

Lifespan:

Circulate for ~120 days; ~3×10^6 are cleaned up by the liver every second.

Pathology:

In sickle cell anemia, erythrocytes deform, reducing blood flow and causing pain and organ damage.


(a) Blood smear (normal)
Neutrophil
Monocyte
Red blood cells (erythrocytes)
Lymphocyte (white blood cell)
50µm
(b) Red blood cells in normal and sickle cell anemia blood smears
Normal
20µm
Sickle cell anemia
Arrows show erythrocytes with a range of abnormal shapes
'Sickle' shaped erythrocyte
20µm

White Blood Cells: Granulocytes

1. **Neutrophils** (most common):
   - Multilobed nucleus (2-5 lobes) with purple-staining granules that release antimicrobial enzymes.
   - Phagocytic; destroy damaged tissue and bacteria, key in inflammation.
2. **Eosinophils**:
   - Bilobed nucleus; quickly exit blood and localize in respiratory and gastrointestinal tissues.
   - Release histaminase and arylsulfatase B to neutralize inflammatory mediators.

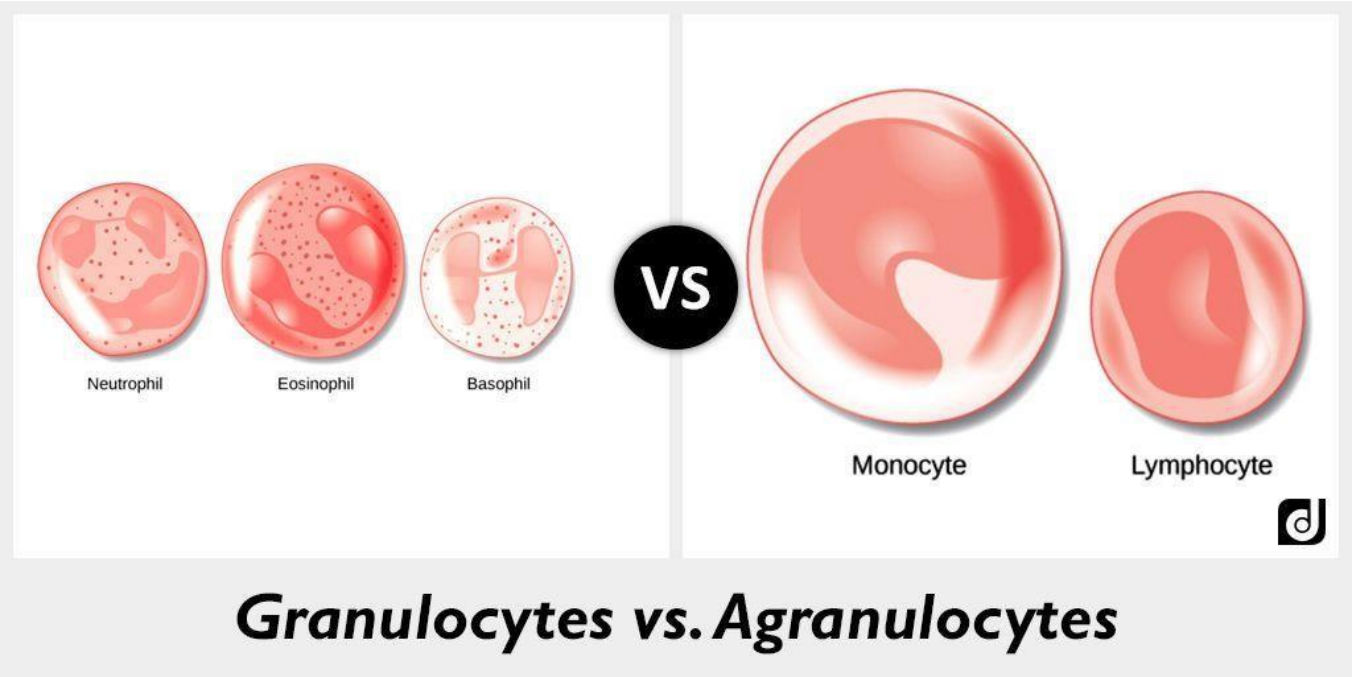
White Blood Cells: Granulocytes

3. **Basophils** (very rare):
   - Contain IgE receptors; involved in allergic reactions and immune responses to parasites.
   - Granules contain histamine, heparin, and other inflammatory mediators.

How They Work Together:

- • **Allergic Reactions:** Basophils and mast cells release histamine to cause inflammation (swelling, redness, itching), while eosinophils arrive later to reduce this inflammation.
- • **Defense Against Parasites:** Eosinophils directly attack parasites, with basophils and mast cells attracting them and other immune cells to the site.

- Together, they balance the immune response, with **basophils** and **mast cells** initiating inflammation and **eosinophils** modulating it.



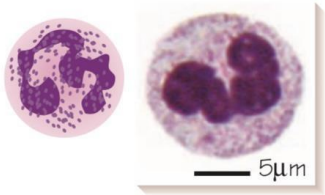
White Blood Cells (Leucocytes)

- White blood cells are less abundant than red blood cells and are classified into **granulocytes** and **agranulocytes.**

(c) Types of white blood cell

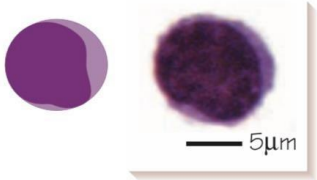
Most common

Neutrophil (granulocyte) 50-70%, 12-14 µm diameter. Multilobed nucleus and pale pink cytoplasm



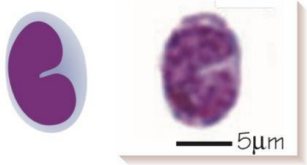
Enter tissues and destroy damaged tissue and bacteria

Lymphocyte (agranulocyte) 20-40%, 7-12 µm diameter. Densely stained nucleus occupies most of the cell, thin rim of cytoplasm



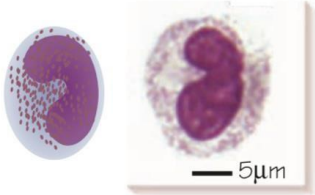
B- lymphocytes, develop into antibody producing cells.
T-lymphocytes recognize foreign antigens, and destroy foreign cells (killer-T cells) or help B cells mount an immune response(helper-T cells)

Monocyte (agranulocyte) 2-10%, up to 20 µm diameter. Bean shaped nucleus, with visible chromatin strands. Grey-blue cytoplasm



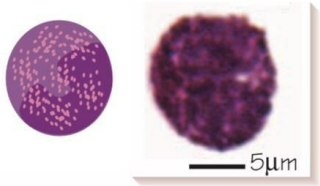
Enter tissues and differentiate into macrophages, which are highly phagocytic, and destroy bacteria and dead tissue

Eosinophil (granulocyte) 1-5%, 12-15 µm diameter Bilobed nucleus. Cytoplasm has lots of large granules



Defends against parasites, and role in triggering bronchial asthma

Basophil (granulocyte) 1%, 14-16 µm diameter. Bi-lobed dark blue/ purple nucleus, and large cytoplasmic granules



Release histamine as part of the allergic reaction

Least common

White Blood Cells: Agranulocytes

1. **Monocytes:**
   - Circulate for 1–3 days, then differentiate into macrophages in tissues, phagocytosing dead cells and bacteria.

2. **Lymphocytes:**
   - **B-cells**: Mature in the bone marrow, producing antibodies.
   - **T-cells**: Mature in the thymus; involved in cell-mediated immunity.

Other Immune Cells

- **Dendritic and Macrophage**: Present antigens to lymphocytes.
- **Mast cells**: Involved in allergic responses; release histamine and heparin.

Platelets (Thrombocytes)

Formation:

- Cytoplasmic fragments from megakaryocytes.

Function:

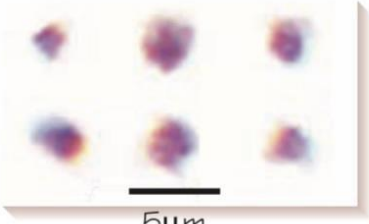
- Form clots, adhere to wound edges, and promote vascular repair.

Structure:

- Lack a nucleus but contain mitochondria, microtubules, and ribosomes.

(d) Platelets

Platelets are 2-4 µm fragments of cytoplasm derived from megakaryocytes



Important for blood clotting, and repair of damaged blood vessels

Aspect	Blood Plasma	Blood Serum
Definition	Liquid component of blood containing clotting factors	Fluid part of blood after clotting has occurred
Contains	Water, electrolytes, proteins, nutrients, hormones, *and clotting factors* (e.g., fibrinogen)	Similar components *except clotting factors*
Obtained by	Centrifuging whole blood with an anticoagulant to prevent clotting	Allowing blood to clot, then centrifuging to remove the clot
Usage	Often used in blood transfusions and testing clotting ability	Commonly used in diagnostic tests due to stable composition

Summary: Plasma includes clotting factors; serum does not.

Lecture

HEMOPOIESIS

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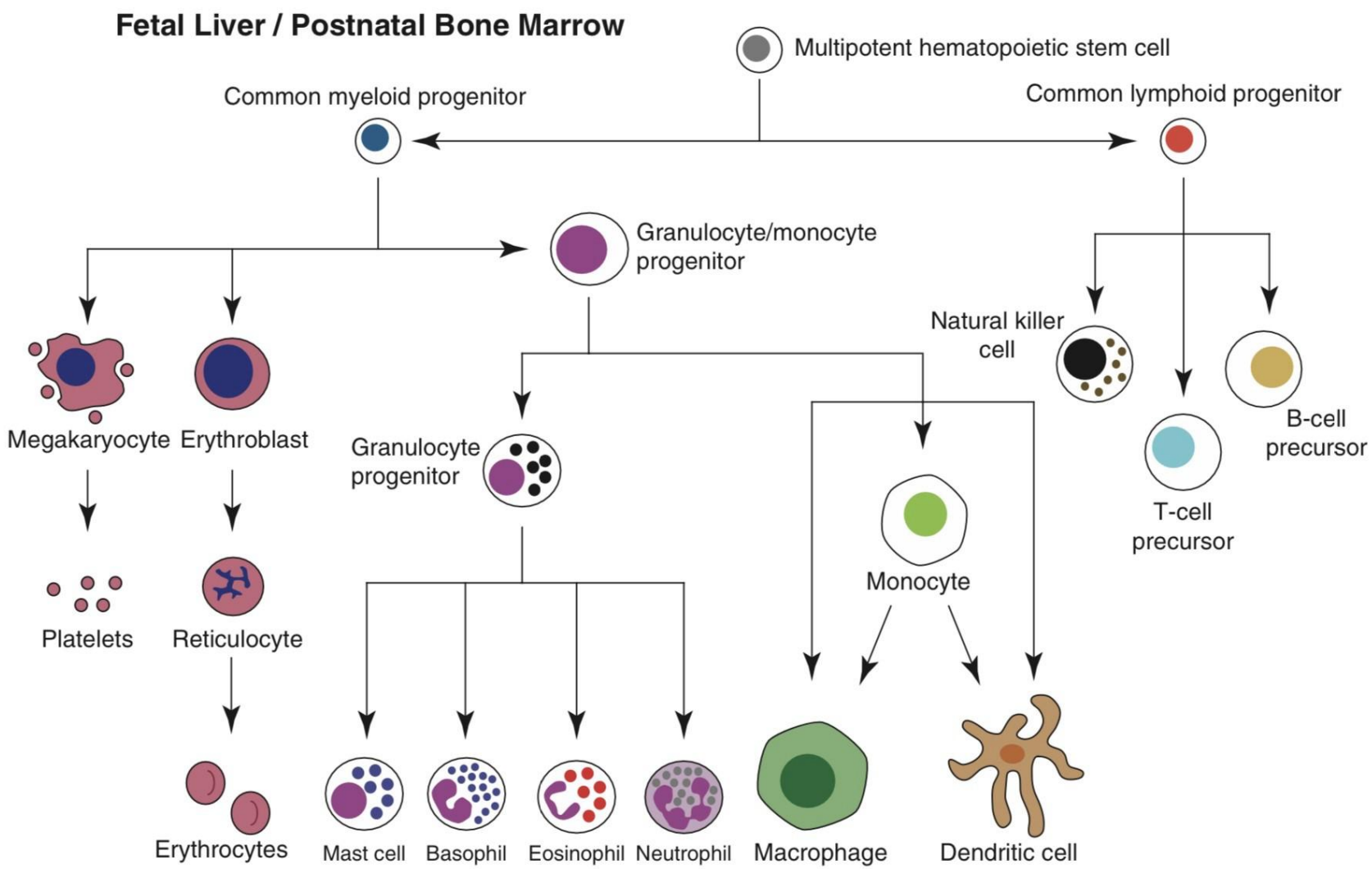
Outlines

- What is Hemopoiesis?
- Types of Stem Cells in Bone Marrow
- Lymphoid Stem Cells
- Myeloid Stem Cells
- Blood Disorders

Hemopoiesis Overview

- **Hemopoiesis** is the continuous process by which mature blood cells develop from precursor cells. It begins in embryonic development and persists throughout adult life, as new blood cells are produced to replace those that age and die.

- All blood cells, including **erythrocytes** (red blood cells), **granulocytes** (neutrophils, eosinophils, and basophils), **agranulocytes** (lymphocytes and monocytes), and **platelets**, originate from pluripotent stem cells within the bone marrow. This tissue resides in the spaces between trabeculae in spongy bone.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Basophil
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Macrophage
Dendritic cell
Natural killer cell
T-cell precursor
B-cell precursor

Bone Marrow Stem Cells

The bone marrow contains **pluripotent stem cells**, which give rise to two main types of multipotent stem cells:

1. **Multipotent Lymphoid Stem Cells:**
   - **T-lymphocytes** (T-cells)
   - **B-lymphocytes** (B-cells)
2. **Multipotent Myeloid Stem Cells**

1. Lymphoid Lineage (Multipotent Lymphoid Stem Cells)

T-lymphocytes (T-cells)

- **Maturation Site**: Thymus
- **Process**: In the thymus, T-cells interact with thymic epithelial cells, preparing them for immune responses.
- **Function**: Mature T-cells enter the bloodstream and travel to peripheral lymphoid tissues (**lymph nodes**, **spleen**). Here, they encounter antigens presented by antigen-presenting cells. T-cells can further differentiate into:
    1. **Helper T-cells (CD4+)**
    2. **Cytotoxic T-cells (CD8+)**

(CD4 and CD8 refer to cluster of differentiation glycoproteins on their surfaces.)

1. Lymphoid Lineage (Multipotent Lymphoid Stem Cells)

B-lymphocytes (B-cells)

- **Maturation Site**: Bone marrow
- **Process**: B-cells begin expressing immunoglobulins (IgM and IgD) on their surface and are tested for binding specificity to ensure they do not react with self-antigens.
- **Function**: Mature B-cells travel via the bloodstream to lymph nodes and other lymphoid tissues, where they can further differentiate into plasma cells that secrete antibodies.

2. Myeloid Lineage (Multipotent Myeloid Stem Cells)

- Myeloid stem cells differentiate into various colony-forming units (CFUs), each giving rise to specific cell types:

2. Myeloid Lineage (Multipotent Myeloid Stem Cells)

Megakaryocyte CFUs

- Develop into **megakaryocytes**, large, multinucleated cells in the bone marrow.
- Function to produce **platelets**, essential for blood clotting.

2. Myeloid Lineage (Multipotent Myeloid Stem Cells)

Erythroid CFUs

- Differentiate into:
- **Erythroblasts,** which gradually shrink and lose their nucleus, forming **reticulocytes**.
- **Reticulocytes** enter the bloodstream and mature into **erythrocytes** within 24 hours.

2. Myeloid Lineage (Multipotent Myeloid Stem Cells)

Granulocyte CFUs

Differentiate into:

- **Neutrophils**: Phagocytic cells that mature through stages like **myelocytes**, which are the last stage capable of cell division. Neutrophils protect against infections.
- **Monocytes**: Circulate in the blood, and upon entering tissues, mature into **macrophages**, which perform phagocytosis and antigen presentation.

2. Myeloid Lineage (Multipotent Myeloid Stem Cells)

Basophil CFUs

Differentiate into:

- **Basophils**, cells involved in allergic reactions by releasing histamine.
- Can further mature into **mast cells** in tissues.

2. Myeloid Lineage (Multipotent Myeloid Stem Cells)

Eosinophil CFUs

Differentiate into:

- **Eosinophils**, which are involved in combating parasitic infections and participate in allergic responses.

❖ **Note**: In general, precursor cells in the bone marrow are larger than their mature counterparts in the bloodstream.

Blood Disorders

- Various blood disorders arise from the abnormal proliferation or function of precursor cells in hemopoiesis:

1. **Chronic Lymphocytic Leukemia (CLL)**
2. **Acute Lymphoblastic Leukemia (ALL)**
3. **Acute Myeloid Leukemia (AML)**
4. **Aplastic Anemia**

Blood Disorders

1. **Chronic Lymphocytic Leukemia (CLL)**

- **Description**: The most common form of leukemia, mainly affecting adults. It involves an increase in immature B-cells, which resemble mature B-lymphocytes.
- **Consequence**: This increase in B-cells can lead to immunocompromise. Normal blood smears contain fewer than 2.5×10^9 lymphocytes per liter, but in CLL, this can rise to over 10×10^9 cells per liter (lymphocytosis).

Blood Disorders

2. Acute Lymphoblastic Leukemia (ALL)

- **Description**: Common in children, it is a malignant disorder of lymphoblastic cells.

- https://www.youtube.com/watch?v=FGv72d_GsZ4Ct=175sCpp=ygUPYmxvb2QgaGlzdG9sb2d5
- The Composition and Function of Blood

Blood Disorders

3. Acute Myeloid Leukemia (AML)

- **Description**: The most common malignant myeloid disorder in adults, affecting various blast cell stages in hemopoiesis.
- **Diagnosis**: Bone marrow smears show elevated myeloblasts (over 30% of all cells).
- **Consequence**: Often fatal due to bone marrow failure.

Blood Disorders

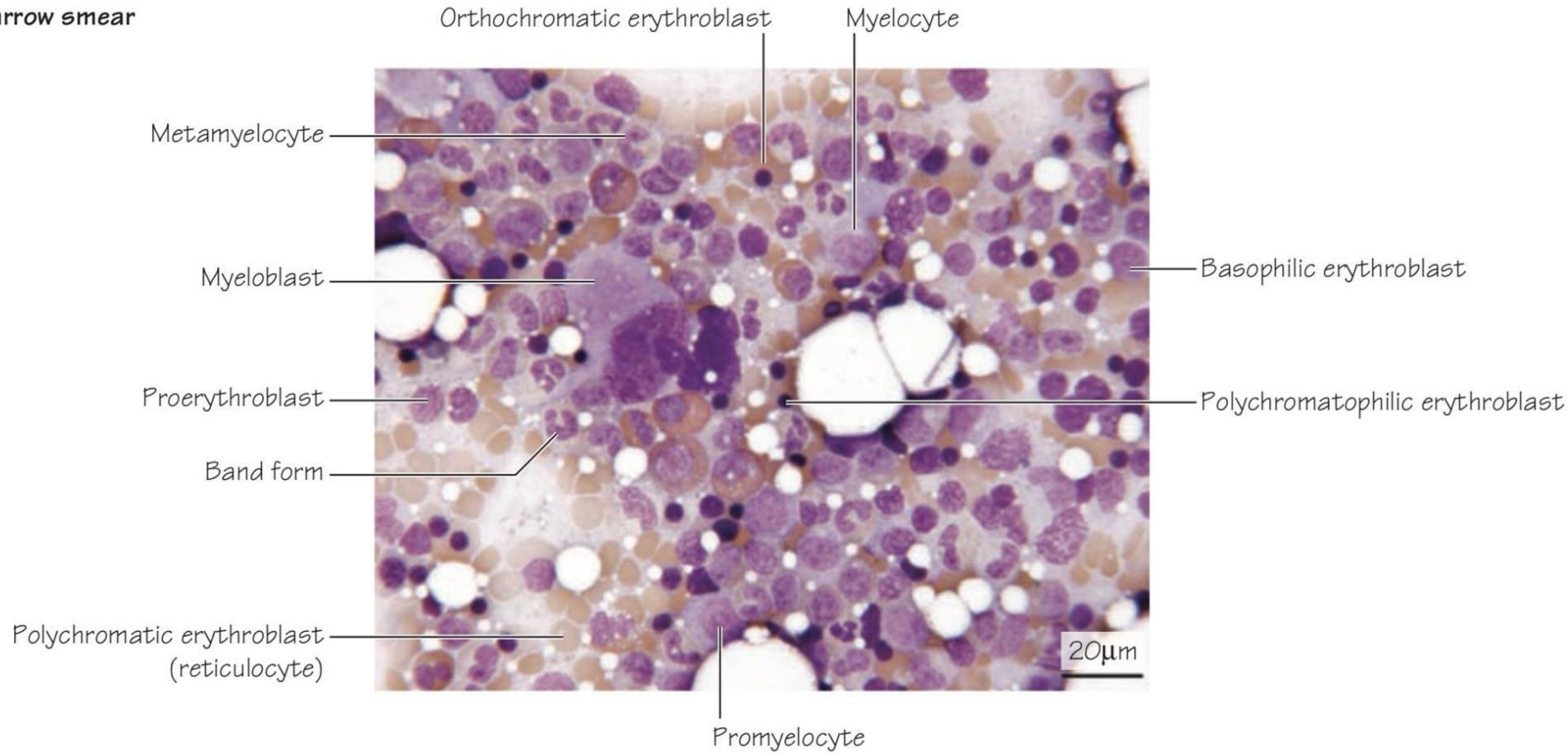
4. Aplastic Anemia

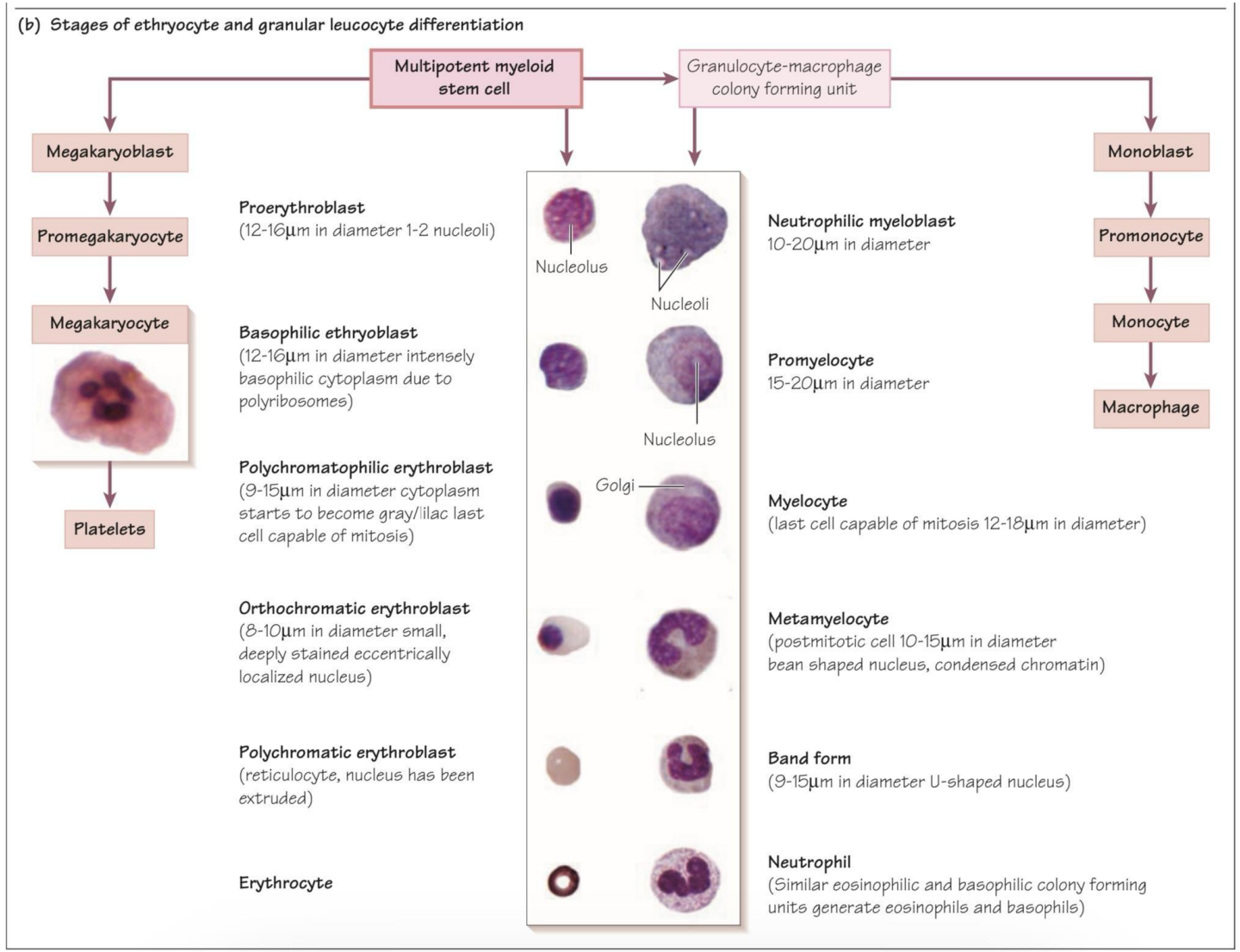
- **Description**: A rare condition in which bone marrow stem cells are destroyed, often by reactive lymphocytes, leading to a reduction in all blood cell types (white cells, red cells, and platelets).
- **Additional Types of Anemia**:

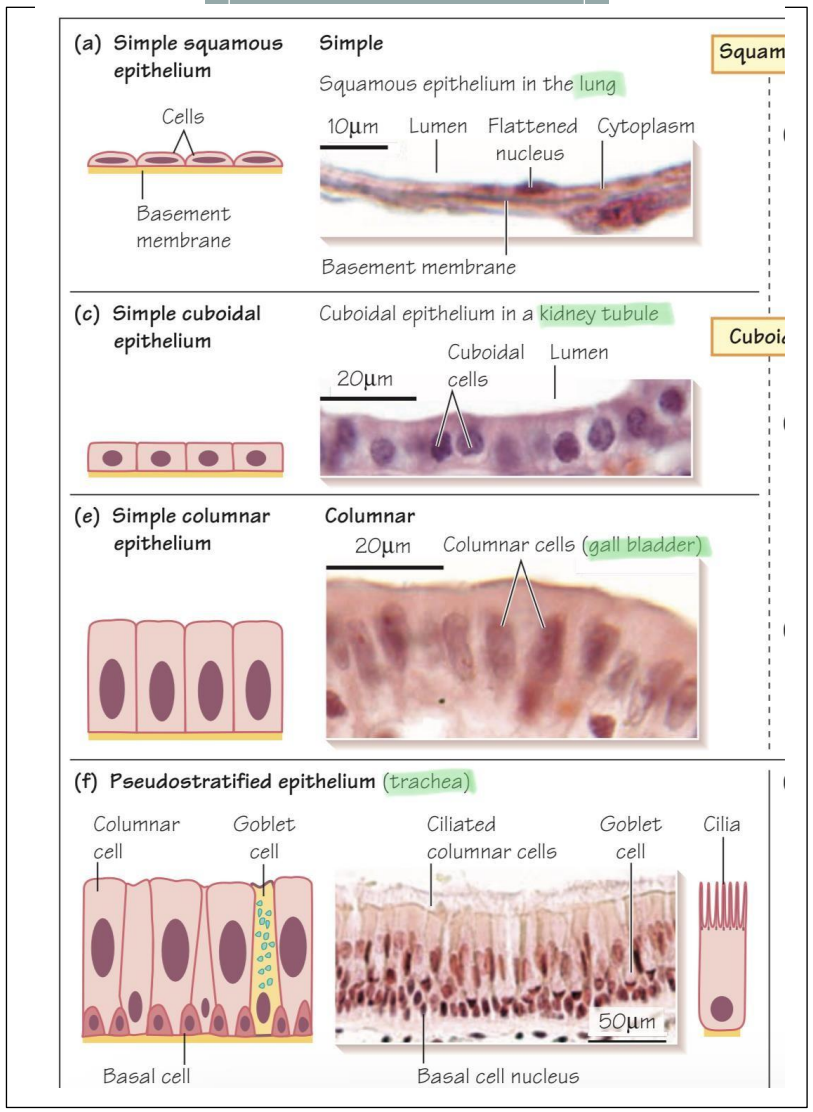
A. **Microcytic Anemia**: Characterized by smaller than normal red blood cells, often due to iron deficiency.

B. **Macrocytic Anemia**: Characterized by larger than normal red blood cells, often due to a deficiency in vitamin B12 or folic acid.

C. **Hemolytic Anemia**: Results from the abnormal breakdown of red blood cells.


(a) Bone marrow smear
Orthochromatic erythroblast
Myelocyte
Metamyelocyte
Myeloblast
Basophilic erythroblast
Proerythroblast
Polychromatophilic erythroblast
Band form
Polychromatic erythroblast
(reticulocyte)
20μm
Promyelocyte


(b) Stages of ethryocyte and granular leucocyte differentiation
Multipotent myeloid stem cell
Granulocyte-macrophage colony forming unit
Megakaryoblast
Promegakaryocyte
Megakaryocyte
Platelets
Proerythroblast
(12-16µm in diameter 1-2 nucleoli)
Basophilic ethryoblast
(12-16µm in diameter intensely basophilic cytoplasm due to polyribosomes)
Polychromatophilic erythroblast
(9-15µm in diameter cytoplasm starts to become gray/ilac last cell capable of mitosis)
Orthochromatic erythroblast
(8-10µm in diameter small, deeply stained eccentrically localized nucleus)
Polychromatic erythroblast
(reticulocyte, nucleus has been extruded)
Erythrocyte
Nucleolus
Nucleoli
Nucleolus
Golgi
Neutrophilic myeloblast
10-20µm in diameter
Promyelocyte
15-20µm in diameter
Myelocyte
(last cell capable of mitosis 12-18µm in diameter)
Metamyelocyte
(postmitotic cell 10-15µm in diameter bean shaped nucleus, condensed chromatin)
Band form
(9-15µm in diameter U-shaped nucleus)
Neutrophil
(Similar eosinophilic and basophilic colony forming units generate eosinophils and basophils)
Monoblast
Promonocyte
Monocyte
Macrophage



Any Questions or Comments?